1. "Auto Cars" is a taxi service in Sri Lanka which has been in operation for several decades, serving the customers in city and urban areas. As a result of the increased use of information technology such as mobile phones and the internet, "Auto Cars" decided to launch a web-based system that enables customers to reserve vehicles conveniently. This also helps the company to gain a competitive advantage.

The main system features of the web-based system required for "Auto Cars" to reserve taxis for customers are:

a. Customers and drivers can use the online system to register. They will receive an email with their username and password once they have registered.
b. Customers can book taxis via the website. When they reserve a vehicle, they will receive an SMS on their phone with the driver's contact information and vehicle details.
c. Customers can rate drivers based on their experiences with them on trips.
d. In addition, the company employs a phone operator who manually reserves drivers from the system. This is the same as a customer reserving a driver, except there is no logged in customer in the application.
e. When the operator makes a taxi reservation, she enters the customer's phone number into the system, and the customer receives an SMS.
f. The driver closest to the customer must be assigned by the application. When a customer or the operator reserves a vehicle, the system should display a list of available drivers. From this list, the customer can then select the best driver.
g. The driver's location is manually (by the driver) entered into the web-based system or via the telephone operator. Similarly, when reserving a taxi, the customer's location is specified (by the customer or through the operator).

Assume that you are the newly appointed system analyst for "Auto Cars" and assigned to work on this project. Produce a professional report including the answers to the following:

Activity 1

Conduct the feasibility study for the web-based system for "Auto Cars" and produce a feasibility report. Further, evaluate the importance of the feasibility criteria used to investigate the feasibility of the proposed system.

① A feasibility study evaluates whether the proposed system is practical and worth implementing. The feasibility of the 'Auto Cars' web-based taxi reservation system can be evaluated using TELOS criteria.

Technical Feasibility: The system requires a web app, database, email service and SMS gateway. These technologies are widely available and easy to implement. Since driver location is manually entered, GPS integration is not needed. Therefore, the system is technically feasible.

Economic Feasibility: Initial costs include system development, hosting, SMS charges and maintenance. However, the system will increase bookings, reduce manual work, improve efficiency and attract more customers due to online access. Since benefits outweigh costs, the system is economically feasible.

Operational Feasibility: Customers can book taxis online while operators can still make manual bookings for non-technical users. Drivers update their location and customers can rate the drivers, improving service quality. Staff may require minimal training. So, it is operationally feasible.

Legal Feasibility: The system stores personal data (phone numbers & emails), so it must comply with data protection and telecommunication regulations. Since no major legal barriers exist, the system is legally feasible.

Schedule Feasibility: The system is moderately complex but can be developed within a few months using standard technologies.

The proposed 'Auto cars' web-based taxi reservation system is technically, economically, operationally, legally and schedule feasible. Therefore, the project should be approved for development.

Here we used the TELOS feasibility criteria, which ensured,

- The system can be built
- The system is profitable
- User acceptance
- prevention of legal issues
- Timely completion

CASE STUDY

Practice Problem

What if the System is NOT feasible?

① A feasibility study determines whether the proposed system should be implemented. After evaluating the 'Auto Cars' web-based taxi reservation system using the TELOS criteria, the project is NOT recommended for implementation.

Technical Feasibility: The system requires continuous internet access, reliable servers, SMS integration, and real-time driver availability tracking. Many drivers may not have smartphones or stable internet connectivity to update their location regularly. This will prevent accurate driver assignment and reduce system reliability. So, the system is not technically feasible.

Economic Feasibility: Drivers and older passengers may resist the system due to lack of their tech knowledge. So, operators will still need to handle most bookings manually. That means the system will not significantly improve performance.

Economic Feasibility: The project involves high development cost and continuous maintenance. But most customers still book taxis through phone calls, so expected online usage is low. Therefore, return on investment will be poor.

Legal Feasibility: Stroing customer phone numbers and personal data requires compliance with data protection regulations. The company currently lacks security infrastructure and policies to protect user data, making the project legally risky.

Schedule Feasibility: Due to lack of technical expertise, and infrastructure setup requirements, the project will take longer than planned, delaying business operations.

In conclusion: The proposed web-based taxi-reservation system is not feasible due to technical limitations, low economic return, operational resistance and legal risks. There-fore, the project should not be implemented at present.

Instead, 'Auto Cars' should:

- Improve existing phone booking system
- Introduce a basic operator-only computer first
- Gradually move to online booking in the future.

2. Angus McIndoe wants to modernize his popular restaurant by adapting it more closely to the preferences of his repeat customers. He is interested in maintaining information such as customers' preferred seating areas, favorite food items, and usual arrival times in order to improve customer satisfaction and increase business performance. Angus believes that keeping track of additional customer preferences may further enhance the quality of service.

   Assume that you are appointed as a **systems analyst** for this project.

   Develop a clear problem definition for Angus McIndoe's proposed customer

[redacted]

   preference management system, identifying the core problem, key stakeholders, and the objectives the system should achieve.

② Customer Preference Management System

Problem Definition

Currently, Angus McIndoe's restaurant does not keep a formal record of repeat customer's preferences such as seating areas, favorite food items, and usual arrival times. Staff depend on memory, which leads to inconsistent service, difficulty recognizing regular customers. This leads to reduced customer satisfaction as the business grows.

Issues

Issues	Weight
1. No formal customer preference records	10
2. Inconsistent customer service	9
3. Difficulty recognizing repeat customers	8
4. Missed personalization opportunities	8
5. Limited business insights	7
6. Risk of losing competitive advantage	7

Key Stakeholders

1. Restaurant Owner (Angus McIndoe)
   ↳ Wants improved customer loyalty and performance
2. Restaurant Staff
   ↳ Will use the system to access customer preference
3. Repeat Customers (expect personalized services)
4. System developers (Responsible for system implementation)

System Objectives

1. Store customer details and visit history.
2. Record preferences (seating, food, arrival time)
3. Allow staff to quickly retrive preferences
4. Support personalized service and improve satisfaction
5. Help increase customer retention and business performance.

Final Statement

Angus McIndoe's restaurant requires a computerized customer preference management system to record and retrieve customer preferences efficiently in order to provide personalized service and improve business performance.

3.

Description	Task	Must Follow	Time (Weeks)
Draw data flow	A	None	5
Draw decision tree	B	A	4
Revise tree	C	B	10
Write up project	D	C, I	4
Organize data dictionary	E	A	7
Do output prototype	F	None	2
Revise output design	G	F	9
Write use cases	H	None	10
Design database	I	H, E, and G	8

Brian F. O'Byrne ("F," he says, stands for "frozen.") owns a frozen food company and wants to develop an information system for tracking shipments to warehouses.

a. Using the data from the table in Figure 3.EX1, draw a Gantt chart to help Brian organize his design project.

GANTT CHART

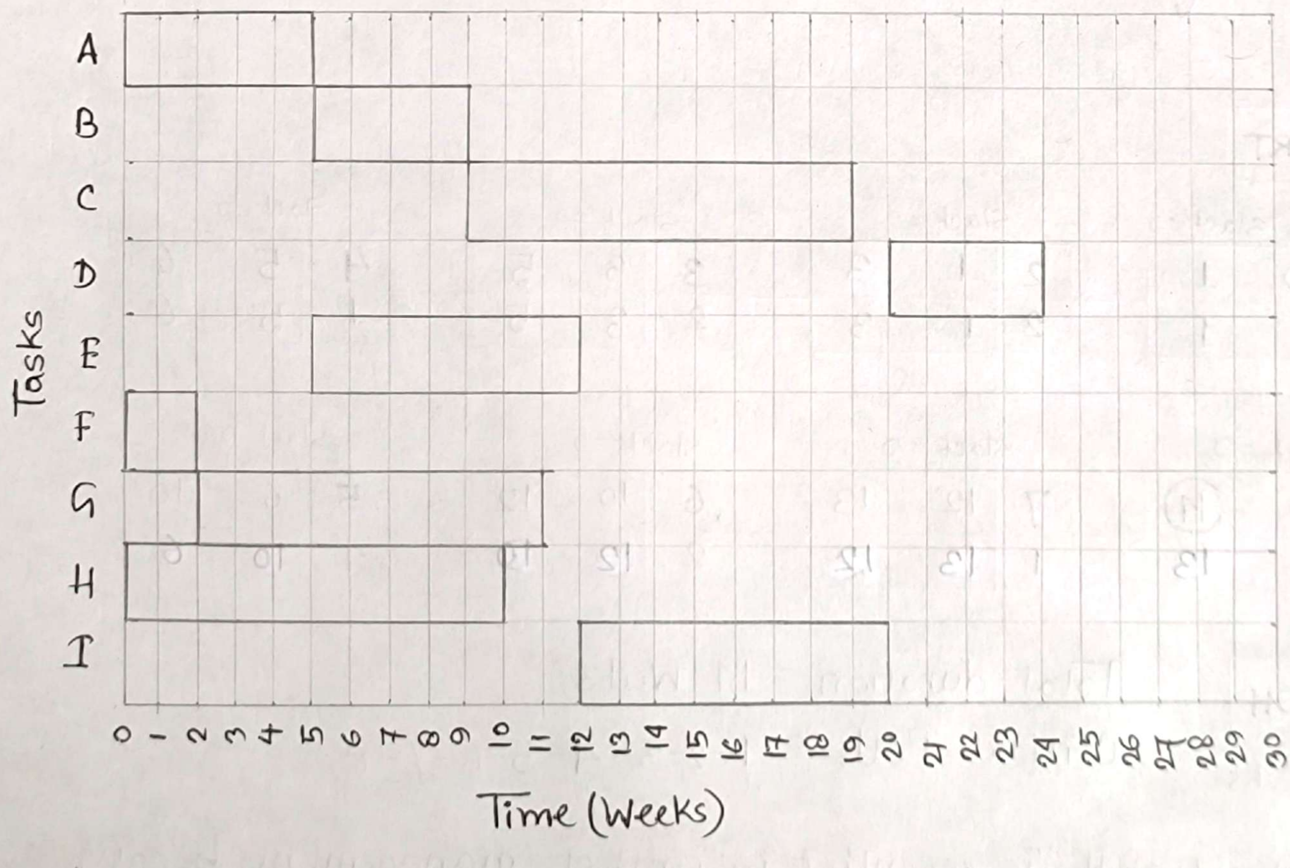
Practice Problem

(3)

A

Description	Task	Must Follow	Time (Weeks)
Draw data flow	A	–	5
Draw decision tree	B	A	4
Revise Tree	C	B	10
Write up project	D	C, I	4
Organize data dictionary	E	A	7
Do output prototype	F	–	2
Revise ouput design	G	F	9
Write use cases	H	–	10
Design databases	I	H, E & G	8

Gantt chart from the given table:



Project Activities – 8 Tasks (Descriptive)

4. **Task 1: Project Planning & Initiation** – Define project objectives, scope, and resources. Duration: **1 week**. Predecessor: None.
5. **Task 2: Requirements Analysis** – Gather and document the system requirements from stakeholders. Duration: **2 weeks**. Predecessor: Task 1.
6. **Task 3: System & Database Design** – Design the system architecture, modules, and database structure. Duration: **2 weeks**. Predecessor: Task 2.
7. **Task 4: User Interface Design** – Create the layout and navigation for the web-based system. Duration: **1 week**. Predecessor: Task 3.
8. **Task 5: System Development / Coding** – Develop and code the system modules. Duration: **4 weeks**. Predecessor: Task 3 & Task 4.
9. **Task 6: Integration & System Testing** – Integrate all modules and perform system testing for errors and performance. Duration: **2 weeks**. Predecessor: Task 5.
10. **Task 7: User Training** – Train users to operate the new system effectively. Duration: **1 week**. Predecessor: Task 6.
11. **Task 8: System Deployment & Review** – Deploy the system, make it operational, and review its performance. Duration: **1 week**. Predecessor: Task 7.

Using the information above, draw a **PERT chart** showing all project activities, their durations, and dependencies. Identify the **critical path** and explain why it is important for project management.

GANTT Chart

Practice Problem

③ B

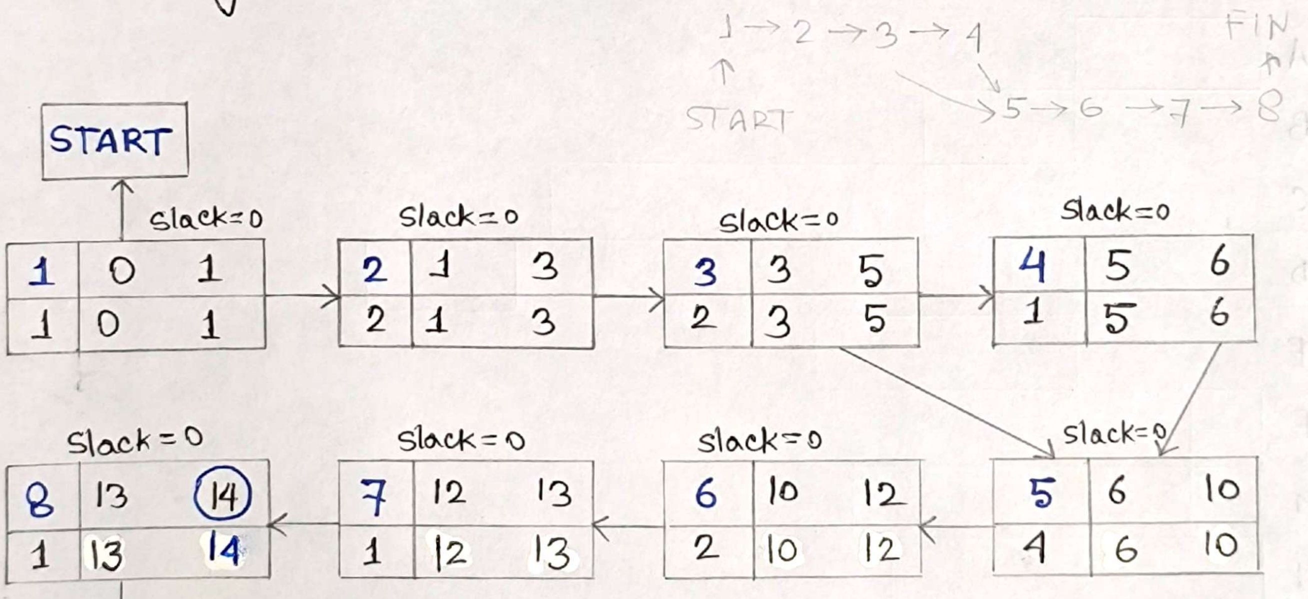
Activity	Task	Predecessor	Duration (Weeks)
Planning & Initiation	1	–	1
Requirement Analysis	2	1	2
System & Database Design	3	2	2
UI Design	4	3	1
System Development	5	3,4	4
Integration & System Testing	6	5	2
User Training	7	6	1
System Deployment & Review	8	7	1

Forward pass ⇒ Maximum

Backward pass ⇒ Minimum

A	ES	EF
t	LS	LF

PERT diagram from the table:



FINISH

14 Weeks

Total duration = 14 Weeks

Critical Path = 1-2-3-4-5-6-7-8

The critical path is essential in project management because it identifies the longest sequence of dependend tasks that determines the shortest possible project during. Tasks with slack time of zero (0) form the critical path.